

10 Turing Machines

- We still have..
 - a set of states
 - some final states
- The input alphabet is a subset of the input alphabet
- We have a symbol for the space: $\square$.
 - Not really different from λ
- This is deterministic
- **You don't have to process all the input**
 - All you have to do is end up in the final state
- Imagine we have a tape, and some symbol a . The read-write head starts as some state q_0 .
 - Now we write over a to a value d , we now are at q_1
- Now suppose the input abb .
 1. Start at a (q_0).
 2. Move to b (q_1)
 3. Move to b (q_1)
- Example $abcd \implies abq_1c \vdash abeq_2d$
 - A way of describing what we go from to.